

1. Write a shell script that reads command line arguments and prints the number of arguments. Given this sequence of words as command line arguments “To the OS Laboratory at Alpha Lab Welcome”, the script should echo back the information that there are 8 arguments. Additionally it should echo back some of the arguments in the following order: “Welcome To the OS Laboratory at Alpha Lab”
2. Write a shell script that asks for the values of 2 variables: `userv1` and `userv2`, from the user and takes in any type (real/integer/character/string). Try to perform the following operations on the variables: (a) add, (b) multiply (c) subtract (d) division (e) printing the variables in reverse order. Print appropriate error message/s if the operation cannot be performed, otherwise print the result: “the sum of ‘userv1’ and ‘userv2’ is”. Execute the shell script as many number of times the user wants. Show all possible combinations of the types of variables and result/s of the corresponding operation/s.
3. Write a shell script a) to find out the total count of files and directories in any directory. b) Also list the file and directory names separately with the respective counts. c) The script should also print out the total size of the files created in the past week.
4. Write a shell script that counts and prints the total number of files in a particular directory and in all of its sub-directories. Also (i) print total no. of files (not directories) in each subdirectory with the name of the subdirectory, (ii) print only those file names that have been created within the past week.
5. Write a shell script that takes 4 file names (Java program files) as command line arguments and prints the frequency of the occurrences of the following 3 strings “public”, “class”, “int” in each file. The output (in tabular format) should clearly denote the frequency of the occurrences of each string in each file.
6. Write a shell script that accepts a file name as an input. The program then asks for a string of characters to be provided by the user. The file will be searched to find whether it contains the given word. (i) If the file contains the given word, the program will display the total number of occurrences of the word. (ii) The program is also required to display the line number/s in which the word has occurred as well the frequency of the word in that line. (Note: the word may occur multiple times in a given line). (iii) If the file does not contain the word, an appropriate message will be displayed. Consider partial match also.
7. Extend the shell script written in question (6) to perform the following task: User is asked to enter another word. (i) The first word (entered in question (6)) to be replaced by the second word, if a match occurs, (ii) Ignore replacing Partial matches, but show that (iii) partial matches do exist, and (iv) match exists, if case sensitivity is ignored
8. Write a script called `birthday_match.sh` that takes two birthdays of the form DD/MM/YYYY (e.g., 15/05/2000) and returns whether there is a match if the two people were born on the same day of the week (e.g., Friday).
9. Write a shell program that finds the factorial of an integer and also mentions the time taken to find out the corresponding factorial. Test the program for different integers.
10. Write a shell program to duplicate the UNIX **rm** command with the following features:
 - a. Instead of deleting the files, it will move them to a **my-deleted-files** directory. If the file already exists in the **my-deleted-files** directory, then the existing file (in the **my-deleted-files**) will have the version number zero (0) appended to it and the newly deleted file will have version number one (1) appended to it, and so on.
 - b. The command will have a switch **-cl** that will clear the entire **my-deleted-files** directory after asking for confirmation.

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